

File Server, Shared Folders, and Access Permissions

Internal IT Knowledge Base / User Guide / SOP Manual

Document Type	Internal IT Knowledge Base / User Guide / SOP
Audience	IT Administrators, IT Support, Help Desk, and Non-IT Staff
Environment	Windows Server
Domain	lab.local
Primary Server	FILESRV01
Related Services	DC01- Active Directory Domain Services (AD DS), DNS

1. Purpose

This document provides a standardized procedure for deploying, configuring, managing, and accessing organizational file shares using a Windows Server file server.

It covers:

- File server configuration
- SMB shared folders
- Share permissions
- NTFS folder permissions
- Active Directory security groups
- Access-Based Enumeration
- Department-specific access
- User access testing
- Common troubleshooting procedures
- Security and administrative best practices

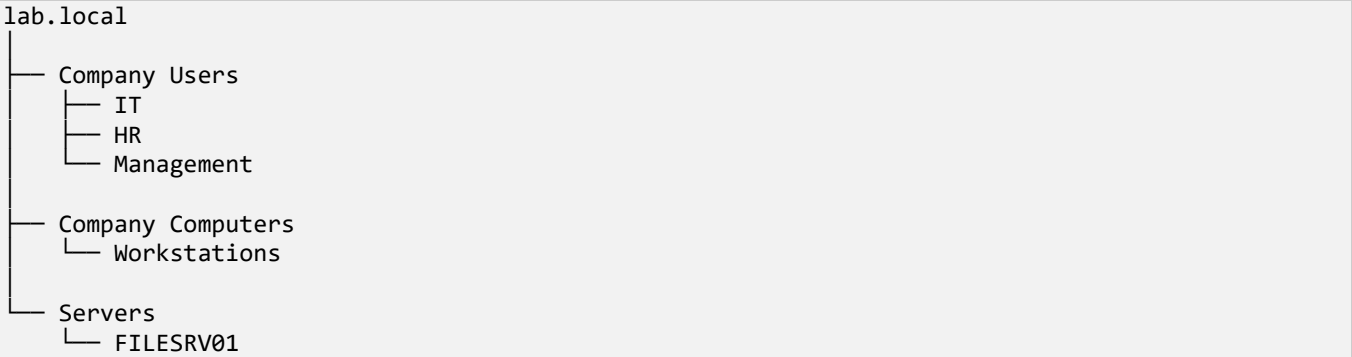
The objective is to ensure that company files are stored centrally and that users only have access to information appropriate to their role.

2. Scope

This procedure applies to:

- Windows Server file servers
- SMB file shares
- Domain-joined Windows computers
- Active Directory users and security groups
- IT administrators managing file permissions
- Employees accessing company shared folders

Example organizational structure:



3. File Server Overview

The organization's file server provides centralized storage for company documents.

Example configuration:

Setting	Value
Server Name	FILESRV01
Domain	lab.local
Server Role	File Server
Protocol	SMB
Example IP	192.168.1.103
DNS Server	192.168.1.100
Data Location	D:\Shares

The file server should be joined to the organization's Active Directory domain.

4. Recommended Folder Structure

The following structure can be used for departmental and general company data:

D:\Shares

- Public
- IT
- HR
- Management

The corresponding network paths are:

\\FILESRV01\Public
\\FILESRV01\IT
\\FILESRV01\HR
\\FILESRV01\Management

Users should normally access files using the server name rather than the server's IP address.

Example:

\\FILESRV01\IT

5. Active Directory Security Groups

Access should be managed using Active Directory security groups rather than assigning permissions individually to every employee.

Recommended groups:

GG_IT
GG_HR
GG_Management

Example:

User
↓
GG_IT
↓
IT Folder

If an employee moves from IT to HR, their membership can be changed without redesigning the folder permissions.

6. File Share Permissions vs NTFS Permissions

Windows file-server security uses two primary permission systems.

6.1 Share Permissions

Share permissions control access when a user connects through SMB.

Example:

```
\\FILESRV01\IT
```

A common administrative model is:

```
Everyone → Full Control
```

The actual restriction is then enforced using NTFS permissions.

6.2 NTFS Permissions

NTFS permissions control access to the actual folders and files stored on the server.

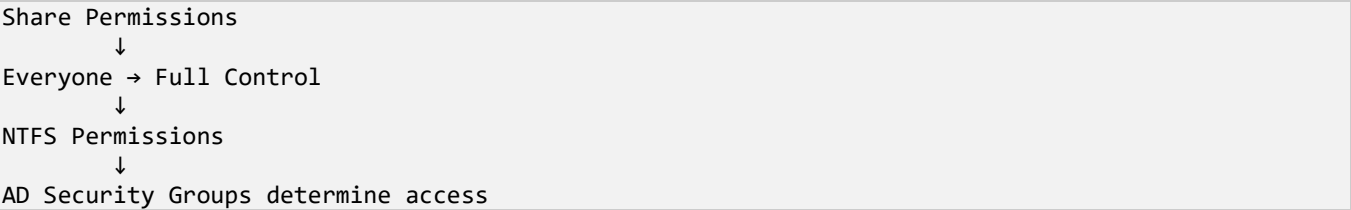
Examples include:

- Full Control
- Modify
- Read & Execute
- List Folder Contents
- Read
- Write

For departmental folders, NTFS permissions should normally be assigned to the appropriate Active Directory security group.

7. Recommended Permission Model

For this environment, use the following general model:



This keeps the share configuration simple while allowing detailed access control through NTFS. The effective access a user receives is determined by the combination of share and NTFS permissions.

8. Public Folder Permissions

The Public folder is intended for documents that should be available to general company users.

Example:

```
D:\Shares\Public
```

Recommended NTFS permissions:

Principal	Permission
Domain Users	Modify
Administrators	Full Control
SYSTEM	Full Control
CREATOR OWNER	Full Control

Example access:

```
\\FILESRV01\Public
```

All authorized domain users may create, modify, and access files in this location.

9. Department Folder Permissions

Department folders should not provide unrestricted access to all employees.

Example:

D:\Shares\IT

Recommended permissions:

Principal	Permission
LAB\GG_IT	Modify
Administrators	Full Control
SYSTEM	Full Control
CREATOR OWNER	Full Control

For HR:

LAB\GG_HR → Modify

For IT:

LAB\GG_IT → Modify

For Management:

LAB\GG_Management → Modify

10. Removing Broad User Permissions

Avoid giving department folders unnecessary permissions through:

BUILTIN\Users

or:

Domain Users

unless all domain users are intentionally supposed to access that folder.

For example, an HR folder should not normally grant:

BUILTIN\Users → Read

because this may allow users outside HR to access HR information.

Instead:

LAB\GG_HR → Modify

should be used.

11. Handling Inherited Permissions

When attempting to remove BUILTIN\Users, Windows may display:

Windows Message: You can't remove Users because this object is inheriting permissions from its parent.

This occurs because the folder is inheriting permissions from its parent directory.

Example:

D:\Shares
↓
D:\Shares\HR
↓
Inherited permissions

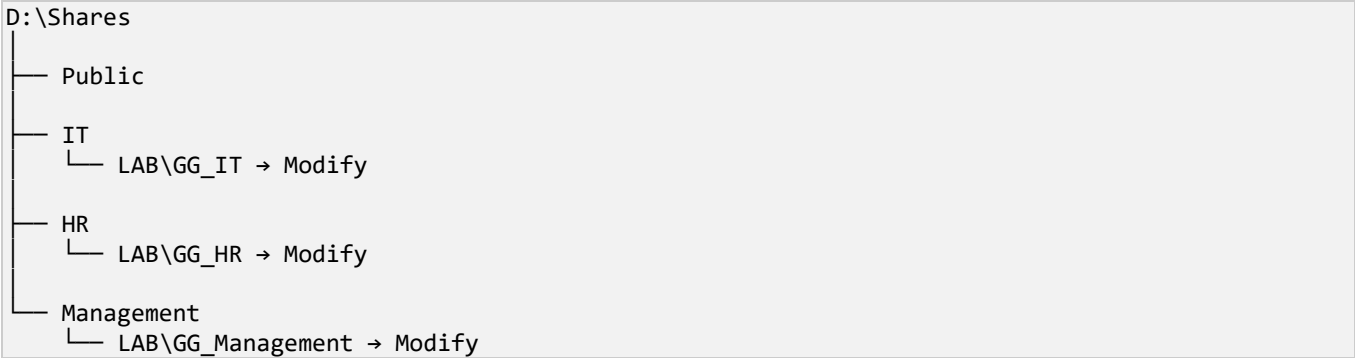
Procedure

1. Right-click the department folder.
2. Select Properties.
3. Select Security.
4. Select Advanced.
5. Locate the inheritance setting.

- 6. Select Disable inheritance.
- 7. When prompted, select: Convert inherited permissions into explicit permissions.
- 8. Select BUILTIN\Users.
- 9. Select Remove.
- 10. Add the appropriate Active Directory security group.
- 11. Assign the required permission.
- 12. Apply the changes.

12. Recommended Department Folder Configuration

Example:



Administrative access should remain available to:

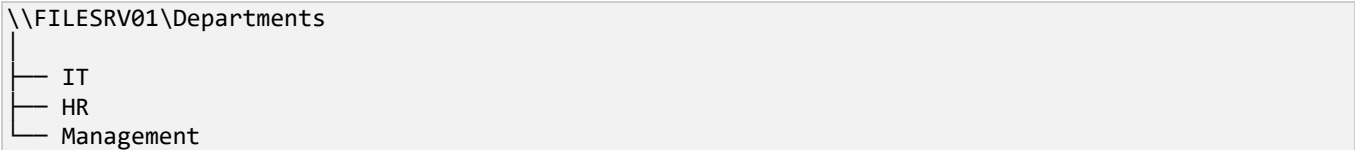
Administrators → Full Control
SYSTEM → Full Control

13. Access-Based Enumeration

Recommended: Enabled

Access-Based Enumeration (ABE) should be enabled for organizational file shares where appropriate. ABE prevents users from seeing files and folders for which they do not have permission.

For example:



An HR employee may only see:

HR

while an IT employee may see:

IT

ABE improves usability and reduces unnecessary exposure of folder names.

Important

Important: Access-Based Enumeration does not replace NTFS permissions. NTFS permissions remain the actual security mechanism.

14. Share Caching

Recommended for this environment: Disabled

The Allow caching of share option is generally unnecessary for a basic office file-server deployment.

Caching is primarily useful for scenarios involving:

- Offline Files
- Mobile users
- Intermittent network connectivity
- Branch-office scenarios

For a normal LAN-based file server, leave this disabled unless there is a documented requirement.

15. Encrypt Data Access

Recommended for the initial lab: Disabled

The Encrypt data access option enables SMB encryption for the share.

It may be appropriate when:

- Sensitive information is being transferred.
- Network security requirements require SMB encryption.
- The organization has a specific compliance requirement.
- File-server traffic crosses less-trusted network segments.

For a basic internal lab, SMB encryption can remain disabled while learning the fundamental permission model.

For a production environment, the security requirements should determine whether SMB encryption is required.

16. Recommended Share Advanced Settings

For the current environment:

Enable access-based enumeration:	YES
Allow caching of share:	NO
Encrypt data access:	NO

These settings can be changed later based on organizational requirements.

17. Creating a New SMB Share

Using Server Manager

1. Open Server Manager.
2. Select File and Storage Services.
3. Select Shares.
4. Select Tasks.
5. Select New Share.
6. Select: SMB Share – Quick
7. Select the appropriate server.
8. Select the folder.
9. Enter the share name.
10. Configure the advanced settings.
11. Review the configuration.
12. Select Create.

Example:

Folder:	D:\Shares\IT
Share name:	IT
Network path:	\\FILESRV01\IT

18. User Access Procedure

For IT Support

Before granting access:

- 1. Confirm the employee's department.
- 2. Confirm the required share.
- 3. Confirm authorization from the appropriate manager if required.
- 4. Add the user to the appropriate AD security group.
- 5. Allow Active Directory replication to complete if necessary.
- 6. Have the user sign out and back in if required.
- 7. Test access.

For End Users

Users should normally access shared folders through:

File Explorer → Address Bar

Example:

\\FILESRV01\Public

or:

\\FILESRV01\IT

Users should not modify permissions on company folders.

19. Mapping a Network Drive

For frequently used shares, IT Support can map a network drive.

Example:

Drive: I:
Location: \\FILESRV01\IT

Users can then access the share from:

File Explorer → This PC

Example:

I: → IT Department Files

Drive mappings can also be deployed automatically through Group Policy.

20. Testing Permissions

Testing should be performed using an actual domain account.

For example:

IT User
↓
Member of GG_IT
↓
Access \\FILESRV01\IT

Expected result:

IT → Access Granted
HR → Access Denied
Management → Access Denied

For an HR user:

HR → Access Granted
IT → Access Denied
Management → Access Denied

21. Permission Testing Matrix

User	Public	IT	HR	Management
IT User	✓	✓	✗	✗
HR User	✓	✗	✓	✗
Management User	✓	✗	✗	✓
Administrator	✓	✓	✓	✓

The exact access model should always follow the organization's approved access-control policy.

22. Troubleshooting

Problem: User cannot access the server

Test connectivity:

```
ping FILESRV01
```

Test DNS:

```
nslookup FILESRV01
```

Test the share:

```
\\FILESRV01
```

If the server cannot be resolved, verify that the client is using the organization's AD DNS server.

Problem: IP address works but server name does not

Example:

```
\\192.168.1.103\IT
```

works, but:

```
\\FILESRV01\IT
```

does not.

This usually indicates a DNS or name-resolution issue.

Verify:

```
nslookup FILESRV01
```

The result should resolve to the file server's IP address.

Problem: User can see a folder but cannot open it

Check:

- 1. NTFS permissions.
- 2. Share permissions.
- 3. AD group membership.
- 4. Group Policy.
- 5. Folder inheritance.
- 6. Explicit deny permissions.
- 7. User logon token/group membership.

Problem: User can access a department folder they should not see

Check:

- 1. BUILTIN\Users
- 2. Domain Users
- 3. Other AD security groups
- 4. Inherited permissions
- 5. Explicit permissions
- 6. Access-Based Enumeration

Remember:

Reminder: ABE hides folders; it does not provide security by itself.

Problem: Permission changes do not appear immediately

Check whether the user's current Windows session has an old group-membership token.

Have the user:

- 1. Sign out.
- 2. Sign back in.
- 3. Test again.

For troubleshooting, check:

```
whoami /groups
```

23. Security Best Practices

IT administrators should follow these principles:

Principle of Least Privilege

Users should receive only the access required to perform their job.

Use Groups

Avoid assigning permissions directly to individual users whenever possible.

Preferred:

```
User → Security Group → Folder
```

Instead of:

```
User → Folder
```

Avoid Broad Permissions

Avoid unnecessary:

```
Everyone → Full Control
```

at the NTFS level.

If Everyone → Full Control is used at the share level, ensure NTFS permissions provide the required restrictions.

Separate Departments

Sensitive departments such as HR and IT should have dedicated security groups and restricted folders.

Review Access Regularly

When an employee:

- Changes department
- Leaves the organization
- Changes role
- Requires additional access

their AD group membership should be reviewed.

24. IT Administrator Checklist

- ☐ Confirm FILESRV01 is joined to lab.local.
- ☐ Confirm static IP configuration.
- ☐ Confirm DNS points to the AD DNS server.
- ☐ Install File Server role.
- ☐ Create D:\Shares.
- ☐ Create department folders.
- ☐ Create AD security groups.
- ☐ Create SMB shares.
- ☐ Configure share permissions.
- ☐ Configure NTFS permissions.
- ☐ Remove unnecessary BUILTIN\Users permissions from restricted folders.

- ☐ Disable inheritance where department-specific permissions require it.
- ☐ Enable Access-Based Enumeration.
- ☐ Leave share caching disabled unless required.
- ☐ Leave SMB encryption disabled unless required by policy.
- ☐ Test access using representative user accounts.
- ☐ Document all exceptions.
- ☐ Review permissions periodically.

25. End-User Quick Guide

How to access company files

Open File Explorer.

Enter the network location provided by IT.

Example:

```
\\FILESRV01\Public
```

For department files:

```
\\FILESRV01\IT
\\FILESRV01\HR
\\FILESRV01\Management
```

If access is denied

Do not attempt to change permissions yourself.

Contact IT Support and provide:

- Your username
- The folder you are trying to access
- The error message
- What you were trying to do

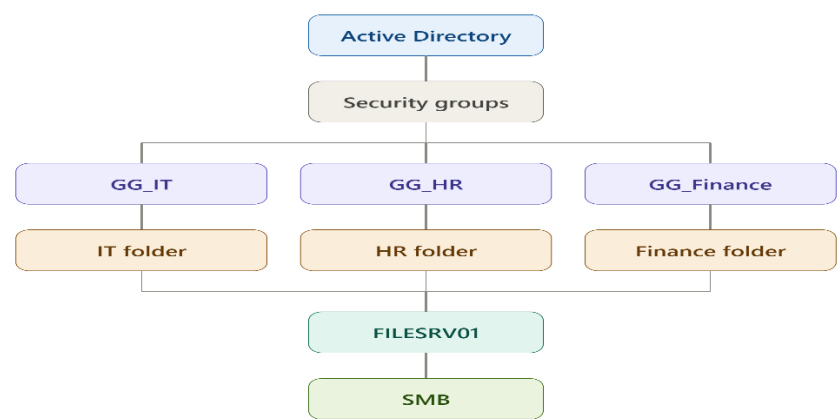
Example:

Example Report: I cannot access \\FILESRV01\IT. Windows reports that access is denied.

IT Support will verify your authorization and group membership.

26. Administrative Design Summary

The recommended design is:



The overall security model is:

```
User
↓
AD Security Group
↓
NTFS Permission
↓
Folder/File
↓
SMB Share
```

27. Recommended Configuration Summary

For the current organization/lab:

Component	Recommended Configuration
File Server	FILESRV01
Domain	lab.local
Protocol	SMB
Storage	D:\Shares
Public Share	\\FILESRV01\Public
IT Share	\\FILESRV01\IT
HR Share	\\FILESRV01\HR
Management Share	\\FILESRV01\Management
Share Permission	Everyone → Full Control
Department Security	AD Security Groups
Department NTFS Access	Modify
Administrators	Full Control
SYSTEM	Full Control
Access-Based Enumeration	Enabled
Share Caching	Disabled
SMB Encryption	Disabled initially
Broad BUILTIN\Users Access	Avoid on restricted folders
Individual User Permissions	Avoid where possible

28. Change Management

Any production permission change should be documented.

Record:

- Requestor
- User affected
- Department
- Folder/share
- Previous access
- New access
- Approver
- Administrator performing the change
- Date and time
- Reason for change

Example:

User:	Employee Name
Department:	IT
Share:	\\FILESRV01\IT
Group Added:	GG_IT
Permission:	Modify
Reason:	Employee joined IT Department
Approved By:	IT Manager
Implemented By:	IT Administrator
Date:	YYYY-MM-DD

29. Final Notes

The file server should be treated as a central company resource, not simply as a folder-sharing computer.

The recommended approach is to:

1. Store data centrally.
2. Use SMB for network access.
3. Use Active Directory groups for authorization.
4. Use NTFS permissions for detailed control.
5. Use broad share permissions only when appropriate.
6. Remove unnecessary broad permissions such as BUILTIN\Users from restricted folders.
7. Use Access-Based Enumeration to improve visibility and usability.
8. Follow least-privilege principles.
9. Test permissions using representative user accounts.
10. Document access changes.

This design provides a strong foundation for expanding the environment later with Group Policy drive mapping, departmental shares, file-server auditing, quotas, backups, DFS, and centralized access management.